

Table 8-37. CHMU Configuration Register (Sheet 2 of 2)

Bit	Attributes	Description
103:96	RW	Down-sampling Factor: If down-sampling is enabled, the device tracks one M2S request over the value encoded in this field. - Bits[99:96]: Down-sampling factor: One of the 16 possible values reported in the Supported Down-sampling Factor field in the CHMU Capability register. - Bits[103:100]: Reserved.
111:104	RW	Reporting Mode: This field is used to enable one of the reporting modes that the device supports. 00h = Enable Epoch-based reporting mode. The device ignores this field if the host programs a value that corresponds to an unsupported capability. 01h = Enable Always-on Reporting Mode. The device ignores this field if the host programs a value that corresponds to an unsupported capability. - All other encodings are Reserved.
127:112	RW	Epoch Length: Host-configured epoch length. - Bits[115:112]: These bits specify the time scale. 1h = 100 us 2h = 1 ms 3h = 10 ms 4h = 100 ms 5h = 1 s - All other encodings are Reserved. - Bits[127:116]: These bits specify the epoch length, using the time scale indicated in bits[115:112].
143:128	RW	Hotlist Notification Threshold: The device generates an interrupt when the number of entries in the Hotlist is greater than or equal to the value in this field and bit[1] of the Overflow Interrupt Status field, "Hotlist Level Crossed", in the CHMU Status register is equal to 0. Upon this condition, bit[1] of the Overflow Interrupt Status field in the CHMU Status register is set to 1. The value in this field cannot exceed the Hotlist size programmed in the Hotlist Size field in the CHMU Capability register. If the value of this field exceeds the programmed Hotlist size, this field is ignored, and no interrupts are generated. When bit[10] of the Flags field, "Interrupt on Hotlist Levels Crossing", is cleared to 0, this field is ignored.
255:144	RsvdP	Reserved

8.2.8.4 CHMU Status [i] (Offset 70h + CHMU Instance Length * i)

The CHMU Status register must be accessed as an 8-byte quantity.

Table 8-38. CHMU Status Register

Bit	Attributes	Description
15:0	RO	Status: This field is used to retrieve CHMU-related status and error information. - Bit[0]: Tracking enabled: When set to 1, the CHMU is enabled. When cleared to 0, the CHMU is disabled. - Bits[15:1]: Reserved.
31:16	RO	Operation In progress: This status information returns the operation currently in progress. 0000h = No operation in progress. Indicates that no operations that can be started through the Control field in the CHMU Configuration register are in progress. 0001h = Enablement in progress. Indicates that enabling of the CHMU is not yet complete. 0002h = Disablement in progress. Indicates that disabling of the CHMU is not yet complete. 0003h = Reset counters in progress. Indicates that resetting of the CHMU counters is not yet complete. - All other encodings are Reserved.
39:32	RO	Counter Width: Current number of bits in each counter used for counting the accesses. This field also indicates the counter width in each Hotlist entry.
47:40	RW1C	Overflow Interrupt Status: - Bit[40]: Hotlist Overflow. When set to 1, the Hotlist has encountered an overflow condition. Additional Hotlist Overflow interrupts are not generated until software clears this bit. - Bit[41]: Hotlist Level Crossed. When set to 1, the Hotlist has achieved the Hotlist Notification Threshold set by the software. Additional Hotlist Level Crossed interrupts are not generated until software clears this bit to 0. - Bits[47:42]: Reserved.
63:48	RsvdP	Reserved

8.2.8.5 CHMU Hotlist Head Register (Offset 78h + CHMU Instance Length * i)

The CHMU Hotlist Head register must be accessed as a 2-byte quantity.

Table 8-39. CHMU Hotlist Head Register

Bit	Attributes	Description
15:0	RW	Head: Index that points to the oldest hot Unit ID in the hotlist that has not yet been read by software. Software reads hot unit entries at this index. This field can be written, by way of the host, to the first unread entry. The Hotlist is considered empty when Head == Tail. The Hotlist is considered full when Head == (Tail + 1) mod (Hotlist Size). In this condition, the number of units in the CHMU Hotlist is equal to the Hotlist Size - 1.

8.2.8.6 CHMU Hotlist Tail Register (Offset 7Ah + CHMU Instance Length * i)

The CHMU Hotlist Tail register must be accessed as a 2-byte quantity. The value of this register must not be changed while the CHMU is enabled. If the value is changed while the CHMU is enabled, the device behavior is undefined.

Table 8-40. CHMU Hotlist Tail Register

Bit	Attributes	Description
15:0	RW	Tail: Index that points to the Hotlist entry in which the device will report the next hot unit. The device increments this field after writing a hot unit entry at this index. The Hotlist is considered empty when Head == Tail. The Hotlist is considered full when Head == (Tail + 1) mod (Hotlist Size). In this condition, the number of units in the CHMU Hotlist is equal to the Hotlist Size - 1.

8.2.8.7 CHMU Range Configuration Bitmap Register (Offset 10h + CHMU Instance Length * i + CHMU Range Configuration Bitmap Offset[i])

This register is used to enable tracking with a 256-MB DPA range granularity. The valid number of Range Configuration Bitmap fields will cover the total device memory capacity. The register may be padded with Reserved bits (with values of 0) until the offset indicated by the device through the CHMU Hotlist Register Offset of the CHMU Capability register. The length of the CHMU Range Configuration Bitmap register is equal to CHMU Hotlist Register Offset - CHMU Range Configuration Bitmap Register Offset. These offset parameters are reported in the CHMU Capability register.

The CHMU Range Configuration Bitmap register must be accessed as an 8-byte quantity.

Table 8-41. CHMU Range Configuration Bitmap Register

Bit	Attributes	Description
63:0	RW	Range Configuration Bitmap 0: Bitmap indicating 256-MB DPA ranges that are enabled to track accesses starting from DPA = 0000 0000 0000 0000h.
127:64	RW	Range Configuration Bitmap 1: Bitmap indicating 256-MB DPA ranges that are enabled to track accesses starting from DPA = 0000 0004 0000 0000h.
...	RW	...

8.2.8.8 CHMU Hotlist Entry Register (Offset 10h + CHMU Instance Length * i + CHMU Hotlist Register Offset[i])

The CHMU Hotlist entries provide the Unit ID that exceeds the programmed hotness threshold and the corresponding counter value. The number of bits for the counter values is the same as the one returned by the device in the CHMU Status register after the unit size is programmed by the Unit Size field in the CHMU Configuration register. The number of bits associated with the Unit IDs depend on the programmed unit size. The counter bits will be placed in the least significant bits of the entry.

Each CHMU Hotlist Entry register must be accessed as an 8-byte quantity.

Table 8-42. CHMU Hotlist Register

Bit	Attributes	Description
63:0	RO	Entry 0: CHMU Hotlist Entry 0, which reports the Unit ID and related counter value that exceeds the programmed Hotness Threshold: • Bits[63:N]: Unit ID • Bits[N-1:0]: Counter value N is the Counter width value returned by the device in the CHMU Status register. The counter value is invalid if Always-on Reporting mode is enabled.
127:64	RO	Entry 1: CHMU Hotlist Entry 1, which reports the Unit ID and related counter value exceeding the programmed Hotness Threshold: • Bits[63:N] Unit ID • Bits[N-1:0] Counter value N is the Counter width value returned by the device in the CHMU Status register. The counter value is invalid if Always-on Reporting mode is enabled.
...	RO	...
(64*M-1):(64*(M-1))	RO	Entry M-1¹: CHMU Hotlist Entry M-1, which reports the Unit ID and related value counter exceeding the programmed Hotness Threshold: • Bits[63:N] Unit ID • Bits[N-1:0] Counter value N is the Counter width value returned by the device in the CHMU Status register. The counter value is Invalid if Always-on Reporting mode is enabled.

1. M is the Hotlist Size in the CHMU Capability register.

8.2.9 CXL Device Register Interface

CXL device registers are mapped in memory space allocated via a standard PCIe BAR. The entry in the Register Locator DVSEC structure (see [Section 8.1.9](#)) with Register Identifier = 03h describes the BAR number and the offset within the BAR where these registers are mapped. The PCIe BAR shall be marked as prefetchable in the PCIe Configuration Space Header. At the beginning of the CXL device register block is a CXL Device Capabilities Array register that defines the size of the CXL Device Capabilities Array followed by a list of CXL Device Capability headers. Each header contains an offset to the capability-specific register structure from the start of the CXL device register block.

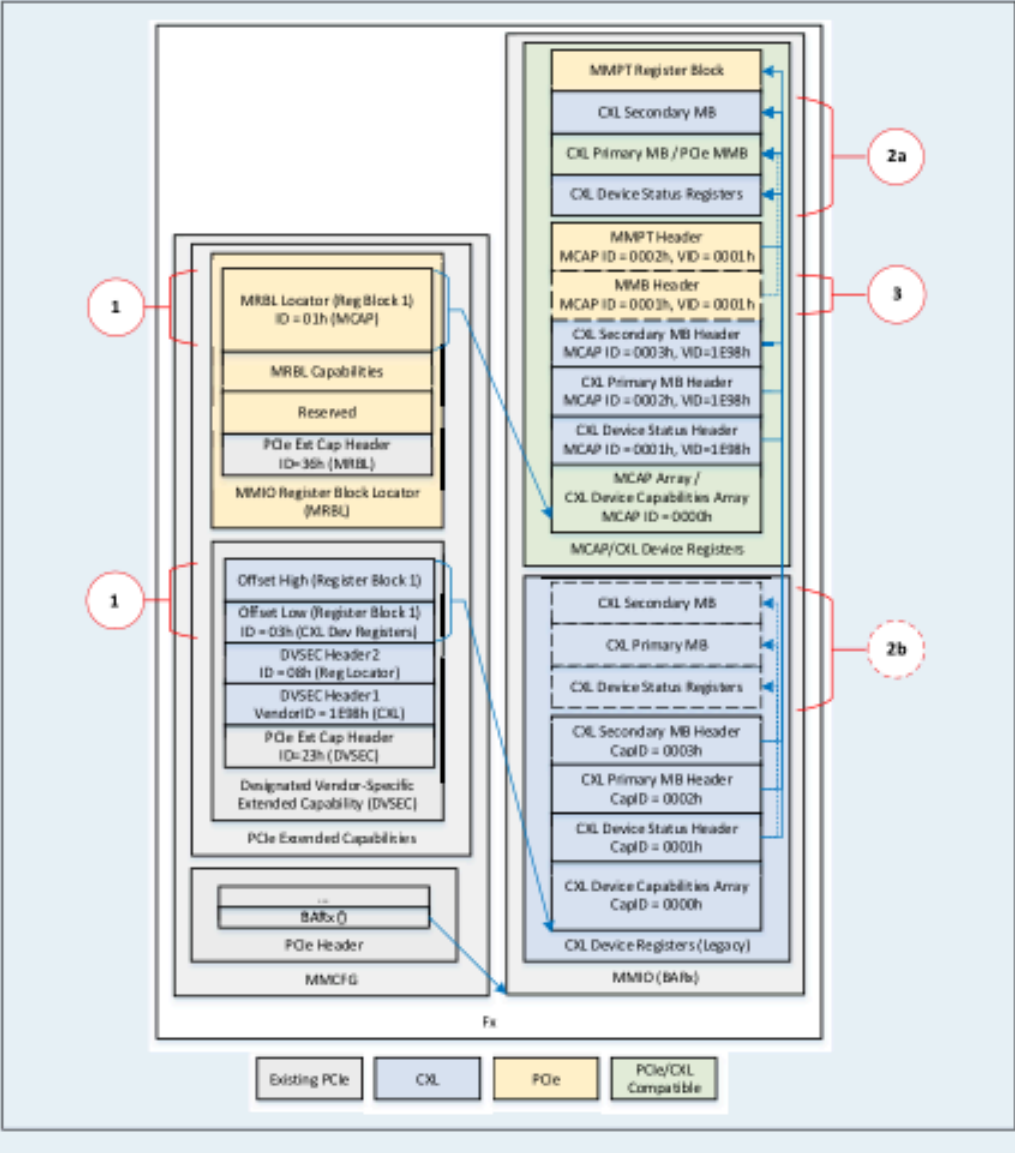
An MLD shall implement one instance of CXL Device registers in the MMIO space of each applicable LD.

IMPLEMENTATION NOTE

Compatibility with PCIe MMIO Capabilities (MCAP) Register Block

CXL components are expected to transition to the PCIe standard MCAP Register Block. CXL defined capabilities. When doing so, CXL components are required to maintain compatibility with legacy CXL software as portrayed in Figure 8-12.

Figure 8-12. PCIe MCAP / CXL Compatibility



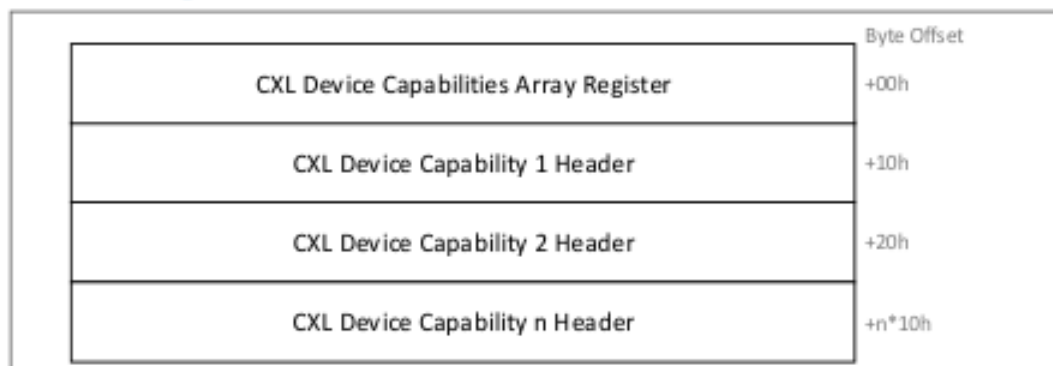
IMPLEMENTATION NOTE**Continued**

1. CXL components are required to advertise CXL defined capabilities in both the legacy CXL Device Capabilities Array and the PCIe MCAP Array. Legacy CXL software discovers the CXL Device Capabilities Array using the entry in the CXL Register Locator DVSEC with Register Identifier = 03h (CXL Device Registers). Updated software discovers the PCIe MCAP Array using the entry in the PCIe MMIO Register Block Locator (MRBL) Extended Capability with Register Block Identifier = 01h (MMIO Capabilities). Updated SW should check the PCIe MCAP Array first and only fallback to the legacy CXL Device Capabilities Array if it's not supported.
2. While CXL defined capabilities must be advertised in both the legacy CXL Device Capabilities Array and the PCIe MCAP Array, CXL components may choose to implement the register structures for CXL defined capabilities in one of the following ways. In either case, legacy software locates the register structure from the offset specified in the header in the legacy CXL Device Capabilities Array. Updated software locates the register structure from the offset specified in the header in the PCIe MCAP Array.
 - a. CXL components may alias the location of CXL defined capabilities specified in the legacy CXL Device Capabilities Array and the PCIe MCAP Array to a single instance of the register structure. In this case, the CXL Device Capabilities Array, the PCIe MCAP Array, and the register structures are required to be located in the same PCIe BAR and both the CXL Device Capabilities Array and the PCIe MCAP Array must be located below the register structures.
 - b. CXL components may choose to implement two instances of the register structures for CXL defined capabilities; one located from the offset specified in the header in the CXL Device Capabilities Array and one located from the offset specified in the header in the PCIe MCAP Array. Updated software is required to use the instance of the register structure located via the PCIe MCAP Array.
3. CXL components that support both the CXL primary mailbox and the PCIe MMB are required to alias the PCIe MMB (Vendor ID = 0001h and MCAP ID = 0001h) to the CXL Primary Mailbox registers. Refer to [Section 9.20](#) for complete details.

No registers defined in [Section 8.2.9](#) are larger than 64-bit wide so that is the maximum access size allowed for these registers. If this rule is not followed, the behavior is undefined.

To illustrate how the fields fit together, the layouts in [Section 8.2.9.1](#), [Section 8.2.9.2](#), and [Figure 8-14](#) (Mailbox registers) are shown as wider than a 64-bit register. Implementations are expected to use any size accesses for this information up to 64 bits without loss of functionality – the information is designed to be accessed in chunks, each no greater than 64 bits.

Figure 8-13. CXL Device Registers



8.2.9.1 CXL Device Capabilities Array Register (Offset 00h)

IMPLEMENTATION NOTE

CXL components are recommended to transition to the equivalent PCIe MCAP Array Register as defined in [PCIe].

Bits	Attributes	Description
15:0	-	Capability ID: See equivalent MCAP ID field in the MCAP Array Register ¹ .
23:16	-	Version: See equivalent MCAP Array Version field in the MCAP Array Register ¹ .
27:24	-	Type: See equivalent MCAP Type field in the MCAP Array Register ¹ . For CXL, bit 27 is fixed to 0 and bits 26:24 are defined in Table 8-43.
31:28	-	Refer to [PCIe] for the definition of this field.
47:32	-	Capabilities Count: See equivalent MCAP Count field in the MCAP Array Register ¹ .
127:48	-	Refer to [PCIe] for the definition of this field.

1. Refer to [PCIe] for the definition of this field.

Table 8-43. CXL Defined Type-Specific Capabilities

Type	Description
0h	Reserved for PCIe ¹ .
1h	Memory Device Capabilities (see Section 8.2.9.5).
2h	FM Mailbox CCI Capabilities (see Section 8.2.9.6).
3-7h	Reserved

1. Refer to [PCIe] for the definition of this type.

8.2.9.2 CXL Device Capability Header Register (Offset: Varies)

Each capability in the CXL device capabilities array is described by a CXL Device Capability Header register that identifies the specific capability and points to the capability register structure in register space.

IMPLEMENTATION NOTE

CXL components are recommended to transition to the equivalent PCIe MCAP Header Register as defined in [PCIe].

Bits	Attributes	Description
15:0	-	Capability ID: See equivalent MCAP ID field in the MCAP Header Register ¹ . See Section 8.2.9.2.1 for the list of CXL defined capability identifiers.
23:16	-	Version: See equivalent MCAP Version field in the MCAP Header Register ¹ .
31:24	-	Refer to [PCIe] for the definition of this field.
63:32	-	Offset: See equivalent MCAP Offset field in the MCAP Header Register ¹ .
95:64	-	Length: See equivalent MCAP Length field in the MCAP Header Register ¹ .
127:96	-	Refer to [PCIe] for the definition of this field.

1. Refer to [PCIe] for the definition of this field.

8.2.9.2.1 CXL Device Capabilities

CXL defined device capability register structures are identified by a 2-byte identifier.

- CXL defined capability identifiers 0000h-3FFFh describe generic CXL device capabilities as specified in Table 8-44.
- Capability identifiers 4000h-7FFFh describe type-specific capabilities associated with the type specified in the CXL Device Capabilities Array register (see Section 8.2.9.1).
- Capability identifiers 8000h-FFFFh describe vendor specific capabilities.

CXL defined Capability identifiers 0000h-3FFFh that are not specified in this table are reserved.

Table 8-44. CXL Defined Capability Identifiers (Vendor ID = 1E98h or 0000h)

Capability ID	Description	Required ¹	Version
0001h	Device Status Registers: Describes the generic CXL device status registers. Only one instance of this register structure shall exist per device.	M	02h
0002h	Primary Mailbox Registers: Describes the primary mailbox registers. Only one instance of this register structure shall exist per device.	M	01h
0003h	Secondary Mailbox Registers: Describes the secondary mailbox registers. At most one instance of this register structure shall exist per device.	O	01h

1. M = Mandatory for all devices that implement the CXL Device Register entry (Register Block Identifier=03h) in the Register Locator DVSEC (see Section 8.1.9). O = Optional.

IMPLEMENTATION NOTE

CXL components are permitted to set the PCIe MCAP Vendor ID to 0000h for CXL defined capabilities. 0000h is a PCI-SIG reserved value for legacy CXL compatibility. However, it is strongly recommended to use the CXL Vendor ID (1E98h) to identify CXL defined capabilities.

8.2.9.3 Device Status Registers (Offset: Varies)

8.2.9.3.1 Event Status Register (Device Status Registers Capability Offset + 00h)

The Event Status register indicates which events are currently ready for host actions, such as fetching event log records. The host may choose to poll for these events by periodically reading this register, or it may choose to enable interrupts for some of these events. The only pollable/interruptible events that are not indicated in this register are mailbox command completions since each set of mailbox registers provides that information.

Unless specified otherwise in the field definitions below, each field is present in version 1 and later of this structure. The device shall report the version of this structure in the Version field of the CXL Device Capability Header register.

Bits	Attributes	Description
31:0	RO	Event Status: When set, one or more event records exist in the specified event log. The device implements a single instance of the event log for host use, which can be accessed either via Primary Mailbox or Secondary mailbox, if applicable. The negotiation between the Operating System and the System Firmware for the ownership of Memory Error Event Logs is managed by CXL_QSC mechanism (See Section 9.18.2). Use the Get and Clear Event Records commands to retrieve and clear the event records. Once the event log has zero event records, the bit is cleared. - Bit[0]: Informational Event Log - Bit[1]: Warning Event Log - Bit[2]: Failure Event Log - Bit[3]: Fatal Event Log - Bit[4]: Dynamic Capacity Event Log¹ - Bits[31:5]: Reserved
63:32	RO	Reserved

1. This bit was introduced with Version=2.

8.2.9.4 Mailbox Registers (Offset: Varies)

CXL defined extensions to the PCIe standard MMIO Mailbox Capability (MMB) Registers are described in this section. Refer to the MMIO Mailbox Capability (MMB) Section of the [PCIe MMPT ECN] for the definition of fields not listed.

There are two types of CXL mailboxes provided through the device's register interface: primary and secondary. Each mailbox represents a unique CCI instance in the device and the properties of each instance are defined in [Section 9.1.1](#). The secondary mailbox does not support background operations. The status of a background operation issued to a device's primary mailbox can be retrieved from the Background Command Status register, as detailed in [Section 8.2.9.4.7](#).

The register interface for both types of mailboxes is the same and is described in the MMB Registers section of [PCIe]. CXL defined extensions to the MMB Registers are called out in this section. The difference between the two types of mailboxes is their intended use and commands allowed. Details on these differences are described in [Section 8.2.9.4.1](#) and [Section 8.2.9.4.2](#).

Commands that require a longer execution time than the MMB command timeout shall be completed asynchronously in the background. Only one command can be executed in the background at a time. The status of a background command can be retrieved from the Background Command Status register. Background commands do not continue to execute across Conventional Resets. For devices with multiple mailboxes, only the primary mailbox shall be used to issue background commands.

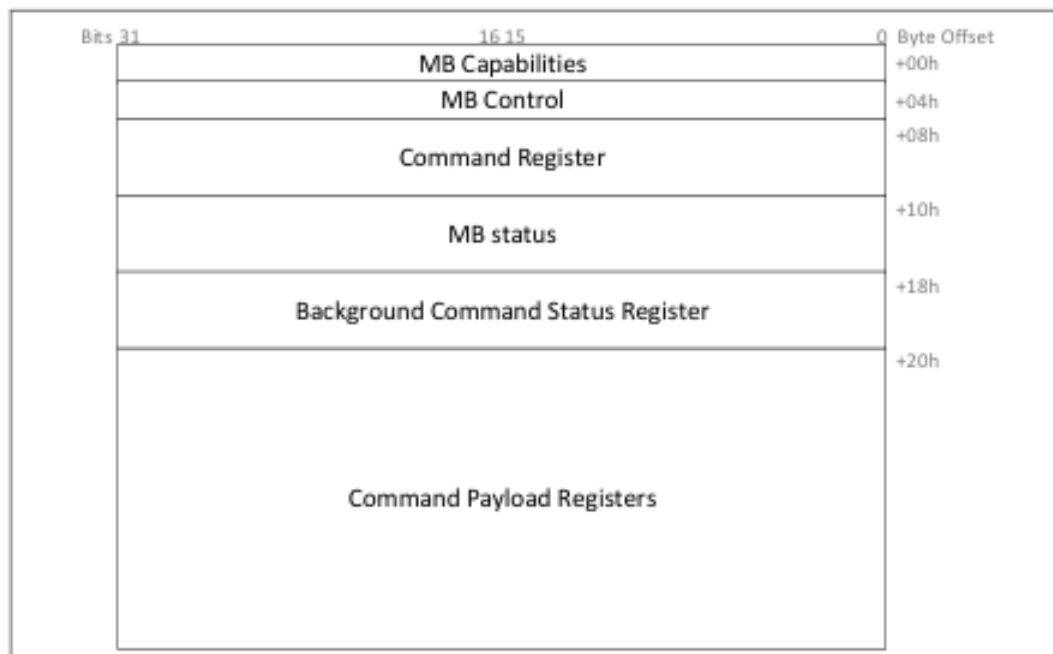
The device shall report the version of these structures in the Version field of the CXL Device Capability Header register.

In case of a timeout, the caller may attempt to recover the device by either issuing CXL or Conventional Reset to the device.

When a command is successfully started as a background operation, the device shall return the Background Command Started return code defined in [Section 8.2.9.4.5.1](#). While the command is executing in the background, the device should update the percentage complete in the Background Command Status register at least once per second. An ongoing background command may be aborted by issuing a Request Abort Background Operation command (see [Section 8.2.10.1.5](#)). It is strongly recommended that devices continue to accept new non-background commands while the background operation is running. The background operation shall not write to the Command Payload registers. Once the command completes in the background, the device shall update the Background Command Status register with the appropriate return code as defined in [Section 8.2.9.4.5.1](#). The caller may then retrieve the results of the background operation from the Background Command Status register.

The Mailbox registers are described in [Figure 8-14](#).

Figure 8-14. Mailbox Registers



8.2.9.4.1 Attributes of the Primary Mailbox

The primary mailbox supports all commands described in [Section 8.2.10](#). The primary mailbox also supports the optional feature to provide mailbox completion interrupts, if implemented by a device. Implementation of the primary mailbox is mandatory.

The exact details on how the primary mailbox is used may vary. The intended use is to provide the main method for submitting commands to the device, used by both pre-boot software and OS software. The platform shall coordinate the use of the primary mailbox so that only one software entity “owns” the mailbox at a given time and that the transfer of ownership happens in-between mailbox commands so that one entity cannot corrupt the mailbox state of the other. The intended practice is that the pre-boot software uses the primary mailbox until control is transferred to the OS being booted, and at that time the OS takes over sole ownership of the primary mailbox until the OS is shut down. Because the physical address of the primary mailbox can change as the result of a PCIe reconfiguration performed by the primary mailbox owner, each time the primary mailbox changes ownership, the new owner shall read the appropriate configuration registers to discover the current location of the mailbox registers, just as it does during device initialization.

8.2.9.4.2 Attributes of the Secondary Mailbox

The secondary mailbox, if implemented by a device, supports only a subset of the commands described in [Section 8.2.10](#). The Command Effects Log shall specify which CXL defined commands are allowed on the secondary mailbox, and all other commands shall return the error Unsupported Mailbox or CCI. The secondary mailbox does not support mailbox completion interrupts. Therefore, any fields related to command interrupts shall be set to 0 on the Secondary MB. Implementation of the secondary mailbox is optional.

The exact details on how the secondary mailbox is used may vary. The intended use is to provide a method for submitting commands to the device by platform firmware that processes events while the OS owns the primary mailbox. By using the secondary mailbox, platform firmware does not corrupt the state of any in-progress mailbox operations on the primary mailbox.

The secondary mailbox shall return identical information as the primary mailbox for a Get Log command issued with Log Identifier=CEL. Devices shall indicate which commands are allowed on the secondary mailbox by setting the Secondary Mailbox Supported flag for the supported opcodes in the Command Effects Log. The set of commands that are supported on the secondary mailbox is implementation specific. It is recommended (but not required) that the secondary mailbox supports all commands in the Events, Logs, and Identify command sets defined in [Section 8.2.10](#).

Since the physical address of the secondary mailbox can change as the result of a PCIe reconfiguration performed by the primary mailbox owner, each time the secondary mailbox is used, the software using it shall read the appropriate configuration registers to discover the current location of the mailbox registers.

8.2.9.4.3 Mailbox Capabilities Register (Mailbox Registers Capability Offset + 00h)

The Mailbox Capabilities Register extends the definition of the PCIe MMB Capabilities Register with the following CXL defined fields.

Bits	Attributes	Description
4:0	-	Payload Size: See equivalent MMB Payload Registers Size field in the MMB Capabilities Register ¹ .
5	-	MB Doorbell Interrupt Capable: See equivalent Command Ready Interrupt Capable field in the MMB Capabilities Register ¹ .

Bits	Attributes	Description
6	RO	Background Command Complete Interrupt Capable: When set, indicates the device supports signaling an MSI/MSI-X interrupt when a command completes in the background. Only valid for the primary mailbox. This bit shall be 0 for the secondary mailbox.
10:7	-	Interrupt Message Number: See equivalent Interrupt Message Number field in the MMB Capabilities Register ¹ .
18:11	-	Mailbox Ready Time: See equivalent MMB Ready Time field in the MMB Capabilities Register ¹ .
22:19	-	Type: See equivalent Type field in the MMB Capabilities Register ¹ . For CXL, bit 22 is fixed to 0 and bits 21:19 are defined in Table 8-45.
31:23	-	Refer to [PCIe] for the definition of this field.

1. Refer to [PCIe] for the definition of this field.

Table 8-45. CXL Defined Mailbox Type Identifiers

Type	Description
0h	Reserved for PCIe ¹ .
1h	Memory Device Commands (see Section 8.2.10.9).
2h	FM API Commands (see Section 8.2.10.10).
3-7h	Reserved

1. Refer to [PCIe] for the definition of this type.

8.2.9.4.4 Mailbox Control Register (Mailbox Registers Capability Offset + 04h)

The Mailbox Control Register extends the definition of the PCIe MMB Control Register with the following CXL defined fields.

Bits	Attributes	Description
0	-	Doorbell: See equivalent Doorbell field in the MMB Control Register ¹ .
1	-	MB Doorbell Interrupt: See equivalent Command Ready Interrupt Enable field in the MMB Control Register ¹ .
2	RW/RO	Background Command Complete Interrupt: If background command complete interrupts are supported on this mailbox, this register is set by the caller to enable signaling an interrupt when the command completes in the background. Ignored if the command is not a background command. RO when the doorbell is set. Ignored if background command complete interrupts are not supported on this mailbox instance (Background Command Complete Interrupt Capable = 0 in the Mailbox Capabilities register). 0 = Disabled 1 = Enabled
31:3	-	Refer to [PCIe] for the definition of this field.

1. Refer to [PCIe] for the definition of this field.

8.2.9.4.5 Command Register (Mailbox Registers Capability Offset + 08h)

The Command Register is equivalent to the definition of the PCIe MMB Control Register with the following CXL defined fields.

Bits	Attributes	Description
15:0	-	Command Opcode: See equivalent MMB Command Opcode field in the MMB Command Register ¹ .
36:16	-	Payload Length: See equivalent MMB Payload Length field in the MMB Command Register ¹ .
63:37	-	Refer to [PCIe] for the definition of this field.

1. Refer to [PCIe] for the definition of this field.

IMPLEMENTATION NOTE

CXL components shall interpret the PCIe MMB Command Opcode Vendor ID = 1E98h or 0000h with CXL defined commands. 0000h is a PCI-SIG reserved value for legacy CXL compatibility. However, it is strongly recommended for callers to use the CXL Vendor ID (1E98h) to identify CXL defined commands.

8.2.9.4.5.1 Command Return Codes

CXL defined command return codes are only interpreted for CXL defined commands.

Table 8-46. CXL defined Command Return Codes (Vendor ID = 1E98h or 0000h) (Sheet 1 of 2)

Value	Definition
0000h	Success: The command successfully completed.
0001h	Background Command Started: The background command successfully started. Refer to the Background Command Status register to retrieve the command result.
0002h	Invalid Input: A command input was invalid.
0003h	Unsupported: The command is not supported.
0004h	Internal Error: The command was not completed because of an internal device error.
0005h	Retry Required: The command was not completed because of a temporary error. An optional single retry may resolve the issue.
0006h	Busy: The device is currently busy processing a background operation. Wait until background command completes and then retry the command.
0007h	Media Disabled: The command could not be completed because it requires media access and media is disabled.
0008h	FW Transfer in Progress: Only one FW package can be transferred at a time. Complete the current FW package transfer before starting a new one.
0009h	FW Transfer Out of Order: The FW package transfer was aborted because the FW package content was transferred out of order.
000Ah	FW Verification Failed: The FW package was not saved to the device because the FW package verification failed.
000Bh	Invalid Slot: The FW slot specified is not supported or not valid for the requested operation.
000Ch	Activation Failed, FW Rolled Back: The new FW failed to activate and rolled back to the previous active FW.
000Dh	Activation Failed, Cold Reset Required: The new FW failed to activate. A cold reset is required.
000Eh	Invalid Handle: One or more Event Record Handles were invalid or specified out of order.

Table 8-46. CXL defined Command Return Codes (Vendor ID = 1E98h or 0000h) (Sheet 2 of 2)

Value	Definition
000Fh	Invalid Physical Address: The physical address specified is invalid.
0010h	Inject Poison Limit Reached: The device's limit on allowed poison injection has been reached. Clear injected poison requests before attempting to inject more.
0011h	Permanent Media Failure: The device could not clear poison because of a permanent issue with the media.
0012h	Aborted: The background command was aborted by the device, either on its own or as a result of a Request Abort Background Operation command.
0013h	Invalid Security State: The command is invalid in the current security state.
0014h	Incorrect Passphrase: The passphrase does not match the currently set passphrase.
0015h	Unsupported Mailbox or CCI: The command is not supported on the mailbox or CCI it was issued on.
0016h	Invalid Payload Length: The input payload length specified for the command is invalid or exceeds the component's Maximum Supported Message Size. The device is required to perform this check prior to processing any command defined in this specification.
0017h	Invalid Log: The log page is not supported or not valid.
0018h	Interrupted: The command could not be successfully completed because of an asynchronous event.
0019h	Unsupported Feature Version: The Feature version in the input payload is not supported.
001Ah	Unsupported Feature Selection Value: The selection value in the input payload is not supported.
001Bh	Feature Transfer in Progress: Only one Feature data can be transferred at a time for each Feature. Complete the current Feature data transfer before starting a new one.
001Ch	Feature Transfer Out of Order: The Feature data transfer was aborted because the Feature data content was transferred out of order.
001Dh	Resources Exhausted: The Device cannot perform the operation because resources are exhausted.
001Eh	Invalid Extent List: The Dynamic Capacity Extent List contains invalid starting DPA and length that are not contained within the Extent List the device is maintaining.
001Fh	Transfer Out of Order: The input parameters data transfer was aborted because it occurred out of order.
0020h	Request Abort Not Supported by Background Operation: The ongoing background operation does not support the Request Abort command.

8.2.9.4.6 Mailbox Status Register (Mailbox Registers Capability Offset + 10h)

The Mailbox Status Register extends the definition of the PCIe MMB Status Register with the following CXL defined fields.

Bits	Attributes	Description
0	RO	Background Operation: When set, the device is executing a command in the background. Only one command can be executing in the background; therefore, additional background commands shall be rejected with the busy return code. Refer to the Background Command Status register to retrieve the status of the background command. Only valid for the primary mailbox. This bit shall be 0 for the secondary mailbox.
31:1	-	Refer to [PCIe] for the definition of this field.
47:32	RO	Return Code: See equivalent MMB Return Code field in the MMB Status Register ¹ .
63:48	RO	Vendor Specific Extended Status: See equivalent Vendor Specific Extended Status field in the MMB Status Register ¹ .

1. Refer to [PCIe] for the definition of this field.

8.2.9.4.7 Background Command Status Register (Mailbox Registers Capability Offset + 18h)

Reports information about the last command executed in the background since the last Conventional Reset. Zeroed if no background command status is available. Valid only for the primary mailbox. This register shall be zeroed on the secondary mailbox.

Bits	Attributes	Description
15:0	RO	Command Opcode: The command identifier of the ongoing command or the last command executed in the background. See Section 8.2.10 for the list of command opcodes.
22:16	RO	Percentage Complete: The percentage complete (0-100) of the background command.
31:23	RsvdP	Reserved
47:32	RO	Return Code: The result of the command run in the background. Only valid when Percentage Complete = 100 or when the Background Operation bit in the Mailbox Status register is cleared to 0. See Section 8.2.9.4.5.1 .
63:48	RO	Vendor Specific Extended Status: The vendor specific extended status of the last background command. Only valid when Percentage Complete = 100.

8.2.9.4.8 Command Payload Registers (Mailbox Registers Capability Offset + 20h)

The Command Payload Registers are equivalent to the definition of the PCIe MMB Payload Registers. Refer to [PCIe] for the definition. See [Section 8.2.10](#) for the format of the payload data for each CXL defined command.

8.2.9.5 Memory Device Capabilities

This section describes the capability registers specific to CXL memory devices that implement the PCIe Configuration Space Header Class Code register defined in [Section 8.1.12.1](#) or advertise Memory Device Capabilities support in the CXL Device Capabilities Array register (see [Section 8.2.9.1](#)).

CXL memory device capability identifiers 4000h-7FFFh that are not specified in [Table 8-47](#) are reserved.

Table 8-47. CXL Defined Memory Device Capabilities Identifiers (Vendor ID = 1E98h or 0000h)

Capability ID	Description	Required ¹	Version
4000h	Memory Device Status Registers: Describes the memory device-specific status registers. Only one instance of this register structure shall exist per device.	M	01h

1. M = mandatory for all CXL memory devices.

IMPLEMENTATION NOTE

CXL components are permitted to set the PCIe MCAP Vendor ID to 0000h for CXL defined memory device capabilities. 0000h is a PCI-SIG reserved value for legacy CXL compatibility. However, it is strongly recommended to use the CXL Vendor ID (1E98h) to identify CXL defined memory device capabilities.

8.2.9.5.1 Memory Device Status Registers (Offset: Varies)

The CXL memory device status registers provide information about the status of the memory device.

8.2.9.5.1.1 Memory Device Status Register (Memory Device Status Registers Capability Offset + 00h)

Unless specified otherwise in the field definitions below, each field is present in version 1 and later of this structure. The device shall report the version of this structure in the Version field of the CXL Device Capability Header register.

Bits	Attributes	Description
0	RO	Device Fatal: When set, the device has encountered a fatal error. Vendor specific device replacement or recovery is recommended.
1	RO	FW Halt: When set, the device has encountered an FW error and is not responding.
3:2	RO	Media Status: Describes the status of the device media. • 00b = Not Ready. Media training is incomplete. • 01b = Ready. The media trained successfully and is ready for use. • 10b = Error. The media failed to train or encountered an error. • 11b = Disabled. Access to the media is disabled. If the media is not in the Ready state, user data is not accessible.
4	RO	Mailbox Interfaces Ready: When set, the device is ready to accept commands through the mailbox register interfaces. Devices that report a nonzero mailbox ready time shall set this bit after a Conventional or CXL reset within the time reported in the Mailbox Capabilities register and it shall remain set until the next reset or the device encounters an error that prevents any mailbox communication.
7:5	RO	Reset Needed: When nonzero, indicates the least impactful reset type needed to return the device to the operational state. A cold reset is considered more impactful than a warm reset. A warm reset is considered more impactful than a hot reset, which is more impactful than a CXL reset. This field returns nonzero value if FW Halt is set, Media Status is in the Error or Disabled state, or the Mailbox Interfaces Ready does not become set. • 000b = Device is operational and a reset is not required • 001b = Cold Reset • 010b = Warm Reset • 011b = Hot Reset • 100b = CXL Reset (device must not report this value if it does not support CXL Reset) • All other encodings are reserved
63:8	RsvdP	Reserved

8.2.9.6 FM Mailbox CCI Capability

This section describes the capability registers specific to FM Mailbox CCIs that implement the PCIe Configuration Space Header Class Code register defined in [Section 8.1.13.1](#) or advertise FM Mailbox CCI Capabilities support in the CXL Device Capabilities Array register (see [Section 8.2.9.1](#)).

FM Mailbox CCI capability identifiers 4000h-7FFFh that are not specified in [Table 8-48](#) are reserved.

Table 8-48. CXL Defined FM Mailbox CCI Capabilities Identifiers (Vendor ID = 1E98h or 0000h)

Capability ID	Description	Required ¹	Version
4000h	FM Mailbox CCI Status Registers: Describes the FM mailbox CCI specific status registers. Only one instance of this register structure shall exist per device.	M	01h

1. M = Mandatory for all FM Mailbox CCI devices.

IMPLEMENTATION NOTE

CXL components are permitted to set the PCIe MCAP Vendor ID to 0000h for CXL defined switch device capabilities. 0000h is a PCI-SIG reserved value for legacy CXL compatibility. However, it is strongly recommended to use the CXL Vendor ID (1E98h) to identify CXL defined switch device capabilities.

8.2.9.6.1 FM Mailbox CCI Status Registers (Offset: Varies)

The FM Mailbox CCI status registers provide information about the status of the FM's Mailbox interface.

8.2.9.6.1.1 FM Mailbox CCI Status Register (FM Mailbox CCI Status Registers Capability Offset + 00h)

Unless specified otherwise in the field definitions below, each field is present in version 1 and later of this structure. The device shall report the version of this structure in the Version field of the CXL Device Capability Header register.

Bits	Attributes	Description
0	RO	Mailbox Interfaces Ready: When set, the device is ready to accept commands through the mailbox register interfaces. Devices that report a nonzero mailbox ready time shall set this bit after a Conventional or CXL reset within the time reported in the Mailbox Capabilities register and it shall remain set until the next reset or the device encounters an error that prevents any mailbox communication.
63:1	RsvdP	Reserved

8.2.10 Component Command Interface

CXL component commands are identified by a 2-byte Opcode and the CXL Vendor ID (1E98h or 0000h).

- Opcodes 0000h-3FFFh describe generic CXL component commands as specified in [Table 8-49](#).
- Opcodes 4000h-BFFFh describe type-specific commands associated with the type specified in the Mailbox Capabilities register (see [Section 8.2.9.4.3](#)). Two Type-Specific command sets are defined - Memory Device command (see [Table 8-141](#)) and FM API commands (see [Table 8-230](#)).
- Opcodes C000h-FFFFh describe vendor specific commands.

Opcodes also provide an implicit *major* version number, which means a command's definition will not change in an incompatible way in future revisions of this specification. Instead, if an incompatible change is required, the specification defining the change will define a new opcode for the changed command. Commands may evolve by defining new fields in the payload definitions that were originally defined as Reserved or by appending new fields, but only in a way where software written using the earlier definition will continue to work correctly, and software written to the new definition can use the 0 value or the payload size to detect components that do not support the new field. This implicit *minor* versioning allows software to be written with the understanding that an opcode will only evolve by adding backward-compatible changes. If a component receives an input payload that is less than the size of the structure it has implemented, but greater than or equal to the Minimum Input Payload Length, then the component shall treat the unsent portion of the structure as 0. For each command, any fields in the input payload that are not included in the calculation

of the Minimum Input Payload Length are explicitly identified. For commands in which no fields are identified, all the fields in the input payload are to be included in the calculation of the Minimum Input Payload Length.

For several commands, the size of the payload that needs to be transferred for a unit of operation may exceed the minimum message size permitted by a CCI. Such commands are defined in such a manner that the payload may be split across multiple messages. For example, the device may implement a mailbox with Payload size of 256 Byte, but its Firmware image may be much larger. The Transfer FW (see [Section 8.2.10.3.2](#)) command is defined such that the host can split the Firmware image in 256 Byte or smaller chunks and transfer each chunk as a separate message over the Mailbox CCI.

Opcodes within the range of 0000h-3FFFh that are not specified in [Table 8-49](#) are reserved.

Generic CXL Device commands can be sent to devices and to switches as long as the devices and switches advertise support for those commands. FM API commands can be sent to switches and to certain types of devices that support them.

Some CCIs may receive commands from more than one command set. The command sets use an overlapping command opcode space, so the transport message contains a field to indicate which command set is being used. For MCTP messages, the DMTF-defined message type field indicates the command set. MCTP Base Specification (see DSP0246) describes a method by which the component advertises which command sets it supports.

Table 8-49. CXL Defined Generic Component Command Opcodes (Vendor ID = 1E98h or 0000h) (Sheet 1 of 2)

Opcode					Required ¹				
					Type 1/2/3 Device		Switch	GFD	
					Mailbox	MCTP		Host Interface ²	FM Interface ³
Command Set Bits[15:8]		Command Bits[7:0]		Combined Opcode					
00h	Information and Status	01h	Identify (Section 8.2.10.1.1)	0001h	PMB	MO	MS	MGFD	MGFD
		02h	Background Operation Status (Section 8.2.10.1.2)	0002h	PMB	MO	MS	MGFD	MGFD
		03h	Get Response Message Limit (Section 8.2.10.1.3)	0003h	PMB	MO	MS	P	MGFD
		04h	Set Response Message Limit (Section 8.2.10.1.4)	0004h	PMB	MO	MS	P	MGFD
		05h	Request Abort Background Operation (Section 8.2.10.1.5)	0005h	O	O	O	P	O
01h	Events	00h	Get Event Records (Section 8.2.10.2.2)	0100h	M	O	O	P	MGFD
		01h	Clear Event Records (Section 8.2.10.2.3)	0101h	M	O	O	P	MGFD
		02h	Get Event Interrupt Policy (Section 8.2.10.2.4)	0102h	M	P	P	P	P
		03h	Set Event Interrupt Policy (Section 8.2.10.2.5)	0103h	M	P	P	P	P
		04h	Get MCTP Event Interrupt Policy (Section 8.2.10.2.6)	0104h	PMB	O	O	P	MGFD
		05h	Set MCTP Event Interrupt Policy (Section 8.2.10.2.7)	0105h	PMB	O	O	P	MGFD
		06h	Event Notification (Section 8.2.10.2.8)	0106h	PMB	O	O	P	MGFD
		07h	GFD Enhanced Event Notification (Section 8.2.10.2.9)	0107h	P	P	P	O	MGFD
		08h	GFD to GAE Enhanced Event Notification (Section 8.2.10.2.10)	0108h	P	P	P	O	MGFD
		09h	Get GAM Buffer (Section 8.2.10.2.11)	0109h	P	P	MGAE	P	P
		0Ah	Set GAM Buffer (Section 8.2.10.2.12)	010Ah	P	P	MGAE	P	P
02h	Firmware Update	00h	Get FW Info (Section 8.2.10.3.1)	0200h	O	O	O	P	O
		01h	Transfer FW (Section 8.2.10.3.2)	0201h	O	O	O	P	O
		02h	Activate FW (Section 8.2.10.3.3)	0202h	O	O	O	P	O

Table 8-49. CXL Defined Generic Component Command Opcodes (Vendor ID = 1E98h or 0000h) (Sheet 2 of 2)

Opcode					Required ¹				
					Type 1/2/3 Device		Switch	GFD	
					Mailbox	MCTP		Host Interface ²	FM Interface ³
Command Set Bits[15:8]	Command Bits[7:0]			Combined Opcode					
03h	Timestamp	00h	Get Timestamp (Section 8.2.10.4.1)	0300h	O	O	O	P	O
		01h	Set Timestamp (Section 8.2.10.4.2)	0301h	O	O	O	P	O
04h	Logs	00h	Get Supported Logs (Section 8.2.10.5.1)	0400h	M	O ⁴	O	MGFD	MGFD
		01h	Get Log (Section 8.2.10.5.2)	0401h	M	O	O	MGFD	MGFD
		02h	Get Log Capabilities (Section 8.2.10.5.3)	0402h	O	O	O	P	O
		03h	Clear Log (Section 8.2.10.5.4)	0403h	O	O	O	P	O
		04h	Populate Log (Section 8.2.10.5.5)	0404h	O	O	O	P	O
		05h	Get Supported Logs Sub-List (Section 8.2.10.5.6)	0405h	O	O ⁴	O	P	O
05h	Features	00h	Get Supported Features (Section 8.2.10.6.1)	0500h	O	O	O	P	O
		01h	Get Feature (Section 8.2.10.6.2)	0501h	O	O	O	P	O
		02h	Set Feature (Section 8.2.10.6.3)	0502h	O	O	O	P	O
06h	Maintenance	00h	Perform Maintenance (Section 8.2.10.7.1)	0600h	O	O	O	P	O
07h	PBR Components	00h	Identify PBR Component (Section 8.2.10.8.1)	0700h	O	MP	MP	P	MGFD
		01h	Claim Ownership (Section 8.2.10.8.2)	0701h	O	MP	MP	P	MGFD
		02h	Read CDAT (Section 8.2.10.8.3)	0702h	MP	MP	MP	MGFD	MGFD

1. M = Mandatory for all devices that implement the CXL Device Register entry (Identifier=03h) in the Register Locator DVSEC (see Section 8.1.9); PMB = prohibited on the primary and secondary mailboxes; P = prohibited; MO = mandatory for components that implement an MCTP-based CCI; MP = Mandatory for PBR Components; MGFD = Mandatory for GFDs; MS = mandatory on MCTP-based CCIs for components that support the FM API MCTP message type; MGAE = Mandatory for GAEs; O = Optional.
2. "Host Interface" refers to commands issued/received via the GFD Proxying mechanism.
3. "FM Interface" refers to commands issued/received via the Fabric Crawl Out mechanism.
4. Type 3 devices implementing support for the Get Supported Logs opcode on an MCTP-based CCI shall also support the Get Supported Logs Sub-List opcode.